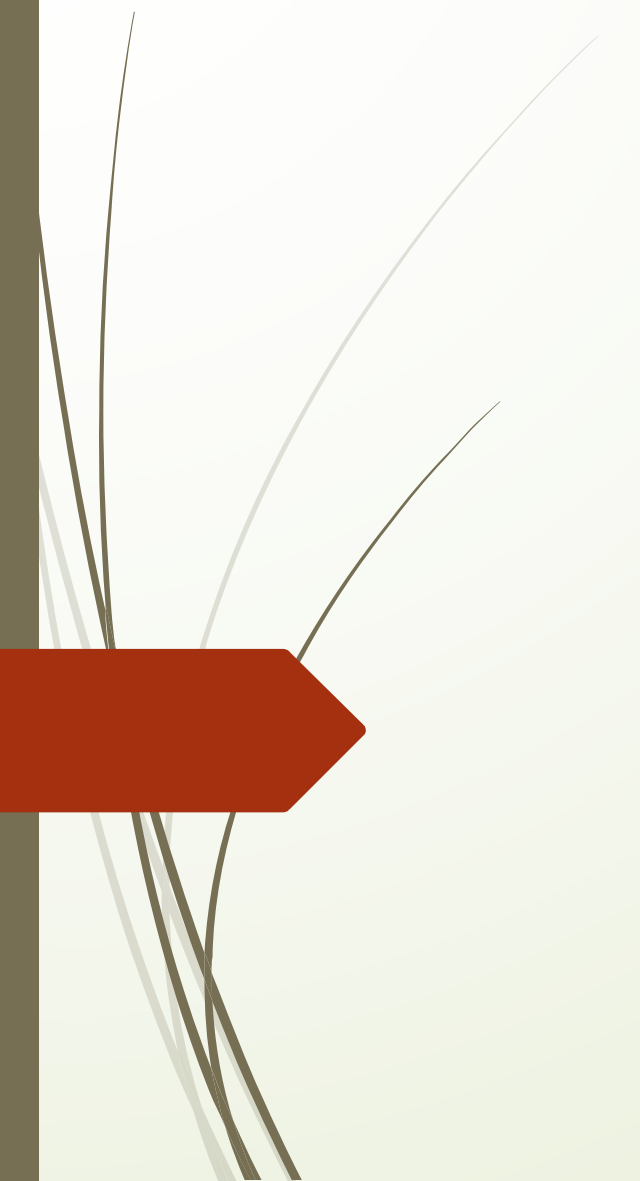


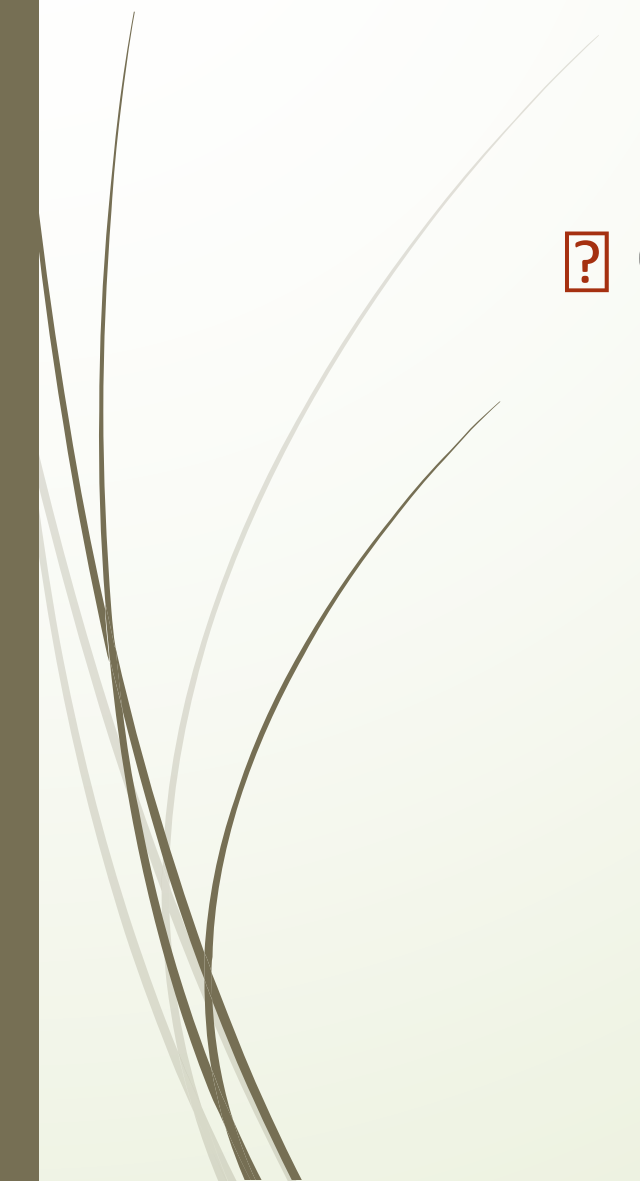


Natural Language Processing

Information Retrieval



Reading



 **Ch. 15 of:** C.D. Manning and H. Schutze, Foundations of Statistical Natural Language Processing, MIT Press, 1999.

Overview

- ❑ Definition of the information retrieval problem
- ❑ Approaches to document indexing
 - ❑ manual approaches
 - ❑ automatic approaches
- ❑ Automated retrieval models
 - ❑ Boolean model
 - ❑ Ranked retrieval methods (e.g. vector space model)
- ❑ Term manipulation
 - ❑ stemming, stop words, term weighting
- ❑ Evaluation

Google search

panthers

[The Official Site of the GMB Nottingham Panthers Ice Hockey Team](#)

GMB **PANTHERS** HEAD COACH MIKE ELLIS IS PLEASED TO ANNOUNCE HE HAS RESIGNED GB INTERNATIONALS DAVID CLARKE AND PAUL MORAN.

www.panthers.co.uk/ - 24k - [Cached](#) - [Similar pages](#)

[Fixtures](#)

[Team - All Time Greats](#)

[Team - Who's Who](#)

[Standings](#)

[More results from panthers.co.uk »](#)

[Panthers.com](#)

Official team site with audio and video clips, team news, depth charts, transactions, statistics, and player profiles.

www.panthers.com/ - 38k - [Cached](#) - [Similar pages](#)

[Black Panthers](#)

The term 'Black **Panther**' is quite often used in connection with large black cats - however there is no one distinct species of wild cat called a Black ...

www.abf90.dial.pipex.com/bco/fact2.htm - 6k - [Cached](#) - [Similar pages](#)

[Panthers' Lair: formerly the Nottingham Panthers Unofficial Guestbook](#)

Kendra Hourd's unofficial Nottingham **Panthers** site featuring a guestbook and chatroom for all **Panther's** fans and friendly interlopers.

www.fortunecity.co.uk/olympia/hornblower/102/ - 15k - [Cached](#) - [Similar pages](#)

Google Search Cont.

panthers Africa

[P.O.V. - A Panther in Africa | PBS](#)

On October 30, 1969, Pete O'Neal, a young Black **Panther** in Kansas City, Missouri, was arrested for transporting a gun across state lines.

www.pbs.org/pov/pov2004/apantherinafrica/ - 23k - [Cached](#) - [Similar pages](#)

[BBC NEWS | Americas | Panther pursues his goals in Africa](#)

Daniel Dickinson meets Pete O'Neal, one of the US's longest-serving political exiles, who now runs community projects in his adopted home of Tanzania.

news.bbc.co.uk/2/hi/americas/3231585.stm - 35k - [Cached](#) - [Similar pages](#)

[BBC SPORT | Football | Africa | When Panthers became Crocodiles](#)

Lesotho step in to save the day after Ghana's opponents Gabon fail to show.

news.bbc.co.uk/sport1/hi/football/africa/2313895.stm - 56k - [Cached](#) - [Similar pages](#)

[[More results from news.bbc.co.uk](#)]

[EcoWorld - Animals - Big Cats](#)

You may want to call these cats black **panthers**, but there's really no such ... for the species found in northwest **Africa**, Iran and parts of the Caucasus. ...

www.ecoworld.com/animals/big_cats_black_leopard.cfm - 25k - [Cached](#) - [Similar pages](#)

Google search Cont.

panthers animal Africa

[Free Pictures of African Animals](#)

Over 100 Beautiful Pictures of **African Animals**. ... of horrible men like Hitler, Stalin, Genghis Kahn along with wolves and foxes, hyenas and **panthers**. ...

huntingsociety.org/Africapics.html - 21k - [Cached](#) - [Similar pages](#)

[* Panther - \(Animals\): Definition](#)

Panther - Online Encyclopedia. ... Of all **Africa's animals** none are better known than the mighty "King of the Beasts" immortalised in many Hollywood movies ...

en.mimi.hu/animals/panther.html - 39k - [Cached](#) - [Similar pages](#)

[wolf Wolf animals/thumbs/wolf.jpg animals/wolf.jpg audio/wolf.mp3 ...](#)

Scientists estimate that only about 80 Florida **panthers** now remain in South Florida, making it one of the most endangered **animals** in the world. ...

www.youwild.org/data/animals.xml?time=8404910742761 - 18k - [Cached](#) - [Similar pages](#)

[Wild Animals: black panthers](#)

You are right in saying that black **panthers** are found in **Africa**, ... I can answer questions about wild mammals, as well as other wild **animals**. ...

en.allexperts.com/q/Wild-Animals-705/black-panthers-2.htm - 20k - [Cached](#) - [Similar pages](#)



Information Retrieval (IR)

- ❑ Given a query, find documents that are “relevant” to the query.
 - ❑ Given: a large, static document collection
 - ❑ Given: an information need (keyword-based query)
 - ❑ Task: find **all** and **only** documents relevant to query
- ❑ Typical IR systems:
 - ❑ Search a set of abstracts
 - ❑ Search newspaper articles
 - ❑ Library search
 - ❑ Search the Web



Issues in IR

- ❑ How can I formulate a query?
 - ❑ query type
- ❑ How are the documents represented?
 - ❑ indexing
- ❑ How does the system find the best-matching document?
 - ❑ retrieval model

Issues in IR contd.

- ❑ How does the system find it efficiently?
- ❑ How are the results presented to me?
 - ❑ unsorted list, ranked list, clusters
- ❑ How do we know whether the system is any good?
 - ❑ evaluation

Indexing

- ❑ The task of finding terms that describe documents well
- ❑ **Manual:**
 - ❑ indexing by humans (using fixed vocabularies)
 - ❑ labor and training intensive
- ❑ **Automatic:**
 - ❑ Term manipulation (certain words count as the same term)
 - ❑ Term weighting (certain terms are more important than others)
 - ❑ Index terms must only derive from text

Manual Indexing

- ❑ Large vocabularies (several thousand items)
 - ❑ MeSH-Medical Subject Headings
 - ❑ ACM-subfields of CS
 - ❑ Library of Congress Subject Headings
- ❑ Advantages:
 - ❑ High precision searches
 - ❑ Work well for closed collections (books in a library)
- ❑ Problems:
 - ❑ Labelers need to be trained to achieve consistency
 - ❑ Documents dynamic □ schemes change constantly

Example: Manual Indexing | MeSH

- ❓ Can 'drill down' from any heading to more and more specific/detailed terms

Eye Diseases [C11]

Asthenopia [C11.093]

➤ Conjunctival Diseases [C11.187]

Conjunctival Neoplasms [C11.187.169]

Conjunctivitis [C11.187.183] +

Pterygium [C11.187.781]

Xerophthalmia [C11.187.810]

Corneal Diseases [C11.204] +

Eye Abnormalities [C11.250] +

Eye Diseases, Hereditary [C11.270] +

Eye Hemorrhage [C11.290] +

Eye Infections [C11.294] +



Automatic Indexing

- ❑ No predefined set of index terms
- ❑ Instead: use natural language as indexing language
- ❑ Words in the document give information about its content
- ❑ Implementation of indexes: inverted files



Inverted files

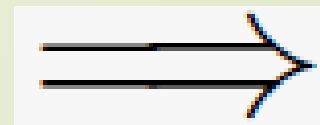
A small collection of documents . . .

<i>Document</i>	<i>Text</i>
1	Pease porridge hot, pease porridge cold
2	Pease porridge in the pot
3	Nine days old
4	Some like it hot, some like it cold
5	Some like it in the pot
6	Nine days old

Inverted files Contd.

- ❑ A basic inverted file index
- ❑ records for each term,
- ❑ the ids of the documents in which it appears
- ❑ only matters if it does or does not appear (not how many times)

<i>Doc</i>	<i>Text</i>
1	Pease porridge hot, pease porridge cold
2	Pease porridge in the pot
3	Nine days old
4	Some like it hot, some like it cold
5	Some like it in the pot
6	Nine days old

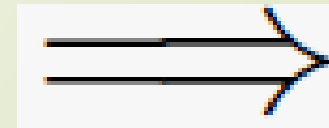


<i>Num</i>	<i>Token</i>	<i>Docs</i>
1	cold	1, 4
2	days	3, 6
3	hot	1, 4
4	in	2, 5
5	it	4, 5
6	like	4, 5
7	nine	3, 6
8	old	3, 6
9	pease	1, 2
10	porridge	1, 2
11	pot	2, 5
12	some	4, 5
13	the	2, 5

Inverted files Contd.

- ❓ A more sophisticated version .
- ❓ also record count of occurrences within each document
- ❓ help find documents more relevant to query

<i>Doc</i>	<i>Text</i>
1	Pease porridge hot, pease porridge cold
2	Pease porridge in the pot
3	Nine days old
4	Some like it hot, some like it cold
5	Some like it in the pot
6	Nine days old

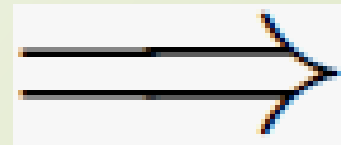


<i>Num</i>	<i>Token</i>	<i>Docs</i>
1	cold	1:1, 4:1
2	days	3:1, 6:1
3	hot	1:1, 4:1
4	in	2:1, 5:1
5	it	4:2, 5:1
6	like	4:2, 5:1
7	nine	3:1, 6:1
8	old	3:1, 6:1
9	pease	1:2, 2:1
10	porridge	1:2, 2:1
11	pot	2:1, 5:1
12	some	4:2, 5:1
13	the	2:1, 5:1

Inverted files Contd.

- ❑ A more sophisticated version . . .
- ❑ also record position of each term occurrence within documents
- ❑ may be useful for searching for **phrases** in documents

Doc	Text
1	Pease porridge hot, pease porridge cold
2	Pease porridge in the pot
3	Nine days old
4	Some like it hot, some like it cold
5	Some like it in the pot
6	Nine days old



Num	Token	Docs
1	cold	1:(6), 4:(8)
2	days	3:(2), 6:(2)
3	hot	1:(3), 4:(4)
4	in	2:(3), 5:(4)
5	it	4:(3, 7), 5:(3)
6	like	4:(2, 6), 5:(2)
7	nine	3:(1), 6:(1)
8	old	3:(3), 6:(3)
9	pease	1:(1, 4), 2:(1)
10	porridge	1:(2, 5), 2:(2)
11	pot	2:(5), 5:(6)
12	some	4:(1, 5), 5:(1)
13	the	2:(4), 5:(5)



Information Retrieval: Methods

- ❑ Boolean search:
 - ❑ Binary decision: is document relevant or not?
 - ❑ Presence of term is necessary and sufficient for match
 - ❑ Boolean operators are set operations (AND, OR)



Information Retrieval: Methods

- ❑ Ranked algorithms:
 - ❑ frequency of document terms
 - ❑ not all search terms necessarily present in document
 - ❑ Incarnations



Bag-of-Words Approach

- ❑ Standard approach to representing documents (and queries) in IR:
- ❑ record what words (terms) present usually, plus count of term in each document
- ❑ Ignores relations between words
- ❑ i.e. of order, proximity, etc.
- ❑ e.g. "May Apple" = "Apple My"



Bag-of-Words Approach

- ❑ Such representations known as “bag of words” approaches
- ❑ c.f. mathematical structure “bag”
- ❑ like a set (i.e. unordered), but records a count for each element

The Vector Space model

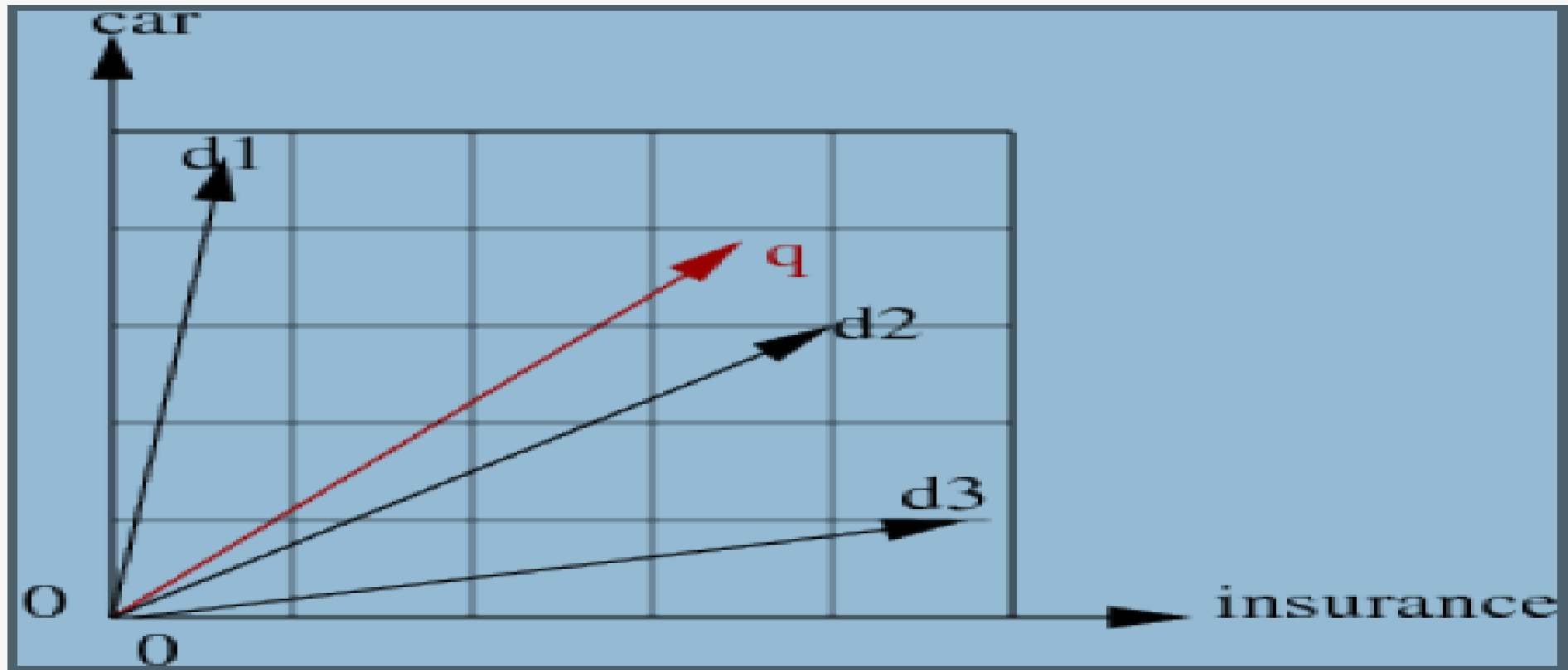
- ❑ Documents are represented as “bags of words”
- ❑ Documents are points in high-dimensional vector space
 - ❑ terms included in index define the dimensions of the space
 - ❑ i.e. each term forms a dimension, so vectors are generally sparse

The Vector Space model

- ❑ Queries are also represented in vector space
 - ❑ based on terms in the query (that are included in the index)
- ❑ Select document(s) with highest document-query similarity
- ❑ Document-query similarity is model for relevance (ranking)

The Vector Space model

Query: car insurance



The Vector Space model

- ❑ Compare **vector of query** against **vector of each document** to find those which are similar

	Term ₁	Term ₂	Term ₃	...	Term _n
Doc ₁	14	6	1	...	0
Doc ₂	0	1	3	...	1
Doc ₃	0	1	0	...	2
...	...	...	...	...	...
Doc _N	4	7	0	...	5
Q	0	1	0	...	1



The Vector Space model



❑ Issues:

- ❑ What is the vector similarity measure?
- ❑ What counts as a term?
- ❑ Are terms weighted?

What is the vector similarity measure? Contd.

- Each document is represented as a vector of n values:

$$\vec{x} = (x_1, x_2, \dots, x_n)$$

- The similarity measure used by vector-space model is the cosine of the angle between two vectors

What is the vector similarity measure? Contd.

The **cosine** of the angle between two vectors $\vec{x}$ and $\vec{y}$ is calculated as:

$$\cos(\vec{x}, \vec{y}) = \frac{\vec{x} \cdot \vec{y}}{|\vec{x}| |\vec{y}|} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}}$$

The cosine value ranges from:

- ❑ 1, for vectors pointing in the same direction, to
- ❑ 0, for orthogonal vectors, to
- ❑ -1, for vectors pointing in opposite directions
- ❑ although, this last case not relevant where all term weights 0

What is the vector similarity measure? Contd.

➡ The (Euclidean) length of vector $\vec{x}$ is computed as:

$$|\vec{x}| = \sqrt{\sum_{i=1}^n x_i^2}$$

❓ c.f. Pythagoras' equation --- a generalization

What is the vector similarity measure? Contd.

❓ Cosine can be interpreted as the normalized correlation coefficient:

$$\cos(\vec{x}, \vec{y}) = \frac{\vec{x} \cdot \vec{y}}{|\vec{x}| |\vec{y}|} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}}$$

i.e. it computes how well the x_i and y_i correlate, and then divides by the length of the vectors, to scale for their magnitude

What is the vector similarity measure? Contd.

- ❑ One option is to **normalize** vectors in advance
- ❑ i.e. scale them so that their **length is equal to 1**
- ❑ then their cosine is equal to the dot product

$$\cos(\vec{x}, \vec{y}) = \vec{x} \cdot \vec{y} = \sum_{i=1}^n x_i y_i$$



What counts as a term?

- ❑ Can use just the words
 - ❑ requires an adequate tokenization approach
- ❑ Capitalization?
 - ❑ normalize all words to lower case or not?
 - ❑ e.g. should we conflate Turkey and turkey ?

What counts as a term?

Contd.

? Single vs. multi-word terms?

? e.g. recognize phrase **cheque book** vs. separate tokens **cheque** and **book**

? Use a stemmer, to conflate morphological variants

CONNECT
CONNECTED
CONNECTING
CONNECTION
CONNECTIONS

WORRY
WORRIED
WORRIES
WORRYING
WORRYINGLY

GALL
GALLING
GALLED
GALLEY
GALLERY

What counts as a term?

Contd.

- ❑ Use a **stop words list**, to exclude 'non-content' words:

a	always	both
about	am	being
above	among	co
across	amongst	could

- ❑ greatly reduces the size of the inverted index
- ❑ but, what if we want to search for phrases that include these terms?



Term Weighting

- ❑ Just record that each term present or not in document
 - ❑ i.e. use binary weights
 - ❑ but document with multiple occurrences of query keyword
 - ❑ may be more relevant
- ❑



Term Weighting Contd.

- ❑ Just use the frequency of term in document
- ❑ But what if the term is generally frequent in collection?
 - ❑ very widely occurring terms not very useful for discriminating relevant documents
- ❑ Want to weight terms highly if they are
 - ❑ frequent in relevant documents . . . but
 - ❑ infrequent in collection as a whole

Term Weighting Contd.

❓ Some key concepts (+ labels for use in equations):

document collection	D	collection (set?) of documents
size of collection	$ D $	total number of documents in collection
term freq	$tf_{w,d}$	number of times w occurs in document d
collection freq	cf_w	number of times w occurs in collection
document freq	df_w	number of documents containing w

The Informativeness of Terms

- ❑ Idea that **less common** terms are **more useful** to finding relevant docs:
- ❑ i.e. that these terms are more **informative**
- ❑ In formativeness: is this idea best addressed using **document frequency** or **collection frequency**?

The Informativeness of Terms Contd.

- ❓ Consider following counts (from New York Times data):

Word	cf_w	df_w
insurance	10440	3997
try	10422	8760

The Informativeness of Terms Cont....

- ❑ term **insurance** semantically focused,
term **try** very general
- ❑ document frequency reflects this
difference
- ❑ collection frequency fails to distinguish
them (i.e. very similar counts)

The Informativeness of Terms Contd.

- ❑ Informativeness is inversely related to (document) frequency
 - ❑ i.e. less common terms are more useful to finding relevant documents
 - ❑ more common terms are less useful to finding relevant documents

The Informativeness of Terms Cont...

- ▶ Perhaps compute measure such as: $\frac{|D|}{df_w}$
 - ▶ value reduces as df_w gets larger, tending to 1 as df_w approaches $|D|$
 - ▶ *but* value v.large for small df_w — *over*-weights such cases
 - ▶ to moderate this effect, can take $\log$

Leads to widely used measure: *inverse document frequency* (idf):

$$idf_{w,D} = \log \frac{|D|}{df_w}$$

Term Weighting: TF.IDF

- ❑ Not all terms describe a document equally well!
- ❑ Putting it all together:
- ❑ Terms which are frequent in a document are better:

$$tf_{w,d} = freq_{w,d}$$

- ❑ Terms that are rare in the document collection are better:

$$idf_{w,D} = \log \frac{|D|}{df_w}$$

- ❑ Combine the two to give TF.IDF term weighting:

$$tf.idf_{w,d,D} = tf_{w,d} \cdot idf_{w,D}$$

Example: TF.IDF

Term	<i>tf</i>	<i>df</i>	$ D $	<i>idf</i>	<i>tf.idf</i>
the	312	28,799	30,000	0.018	5.54
in	179	26,452	30,000	0.055	9.78
general	136	179	30,000	2.224	302.50
fact	131	231	30,000	2.114	276.87
explosives	63	98	30,000	2.486	156.61
nations	45	142	30,000	2.325	104.62
haven	37	227	30,000	2.121	78.48

the:

$$IDF(the) = \log_{10}\left(\frac{30,000}{28,799}\right) = 0.018$$

$$TF.IDF(the) = 312 \cdot 0.018 = 5.54$$

Example: VSM, TF.IDF, cosine

Table of tf.idf values for words in two documents D7655 and D454, and in a query Q (“hunter gatherer Scandinavia”):

$$\Rightarrow \text{sim}(\vec{q}, \vec{d}) = \cos(\vec{q}, \vec{d}) = \frac{\sum_{i=1}^n q_i d_i}{\sqrt{\sum_{i=1}^n q_i^2} \sqrt{\sum_{i=1}^n d_i^2}}$$

	Q	D ₇₆₅₅	D ₄₅₄
hunter	19.2	56.4	112.2
gatherer	34.5	122.4	0
Scandinavia	13.9	0	30.9
30,000	0	457.2	0
years	0	12.4	0
BC	0	200.2	0
prehistoric	0	45.3	0
deer	0	0	23.6
rifle	0	0	452.2
Mesolithic	0	344.2	0
$\sqrt{\sum_{i=1}^n x_i^2}$	41.9	622.9	467.5

Example: VSM, TF.IDF, cosine

$$\Rightarrow \quad \text{sim}(\vec{q}, \vec{d}) = \cos(\vec{q}, \vec{d}) = \frac{\sum_{i=1}^n q_i d_i}{\sqrt{\sum_{i=1}^n q_i^2} \sqrt{\sum_{i=1}^n d_i^2}}$$

$$\begin{aligned} \cos(Q, D_{7655}) &= \frac{(19.2 * 56.4) + (34.5 * 122.4) + \dots + (0 * 0) + (0 * 344.2)}{41.9 * 622.9} \\ &= \frac{5305.68}{26071.72} \\ &= 0.20 \end{aligned}$$

$$\begin{aligned} \cos(Q, D_{454}) &= \frac{(19.2 * 112.2) + (34.5 * 0) + \dots + (0.0 * 452.2) + (0.0 * 0.0)}{41.9 * 467.5} \\ &= \frac{2583.8}{19570.0} \\ &= 0.13 \end{aligned}$$

➡ so document D_{7655} is more similar to Q than D_{454}

What is the vector similarity measure? Contd.

- ❓ Specializing the equation to comparing a query q and document d :

$$\text{sim}(\vec{q}, \vec{d}) = \cos(\vec{q}, \vec{d}) = \frac{\sum_{i=1}^n q_i d_i}{\sqrt{\sum_{i=1}^n q_i^2} \sqrt{\sum_{i=1}^n d_i^2}}$$

- ❓ i.e. computes how well occurrences of each term t correlates in query and document, then scales for the magnitude of the overall vectors